Industrial Training Report Technocolabs Softwares

Hotel Reservation Data Analysis Using SQL

II301

Industrial Internship

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Presented by

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Project Overview

- Analyzed hotel reservation dataset using SQL to uncover trends & operational insights.
- Performed data extraction, cleaning, transformation, and exploratory analysis.
- Computed key parameters like, avg. room price, meal plans, room type demand, and market segments.
- Generated actionable insights to help hotels optimize pricing, improve occupancy, and enhance operational decision-making.

Dataset Overview

- The dataset contains 700+ hotel reservation records, each representing a unique booking.
- Attributes include:-
  - Guest details like no. of citizens in family, booking ID, etc.
  - Duration of stay and Room information
  - Meal Preference based on their availability
  - Date of arrival
  - Mode of booking and its status
  - Pricing of rooms booked
- Overall, the dataset provided a clean and structured view of multiple records that a customer checks before booking, ideal for SQL analysis

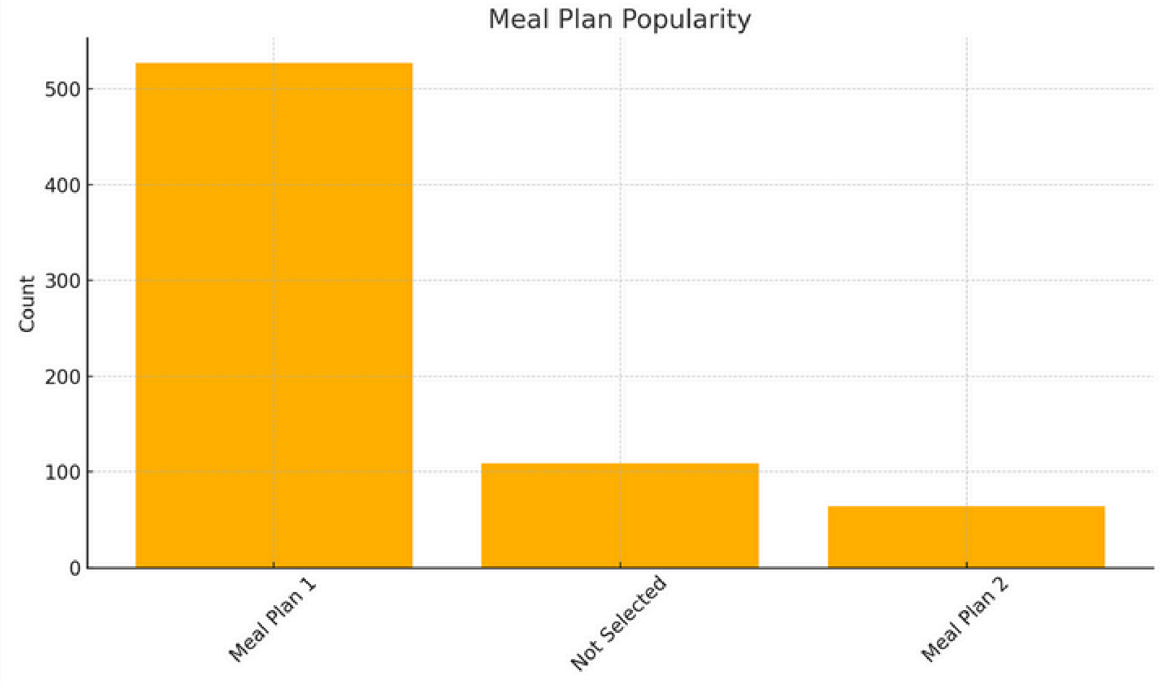
Exploratory Data Analysis

1. Meal Plan Popularity Analysis:-

```sql
#Q2
Select type_of_meal_plan ,
count(Booking_ID)
from hotel.hotel_reservation_dataset
group by type_of_meal_plan;
```

Result Grid | Filter Rows: | Export: | Wi

type_of_meal_plan	count(Booking_ID)
Meal Plan 1	527
Not Selected	109
Meal Plan 2	64



Insights from the chart:-

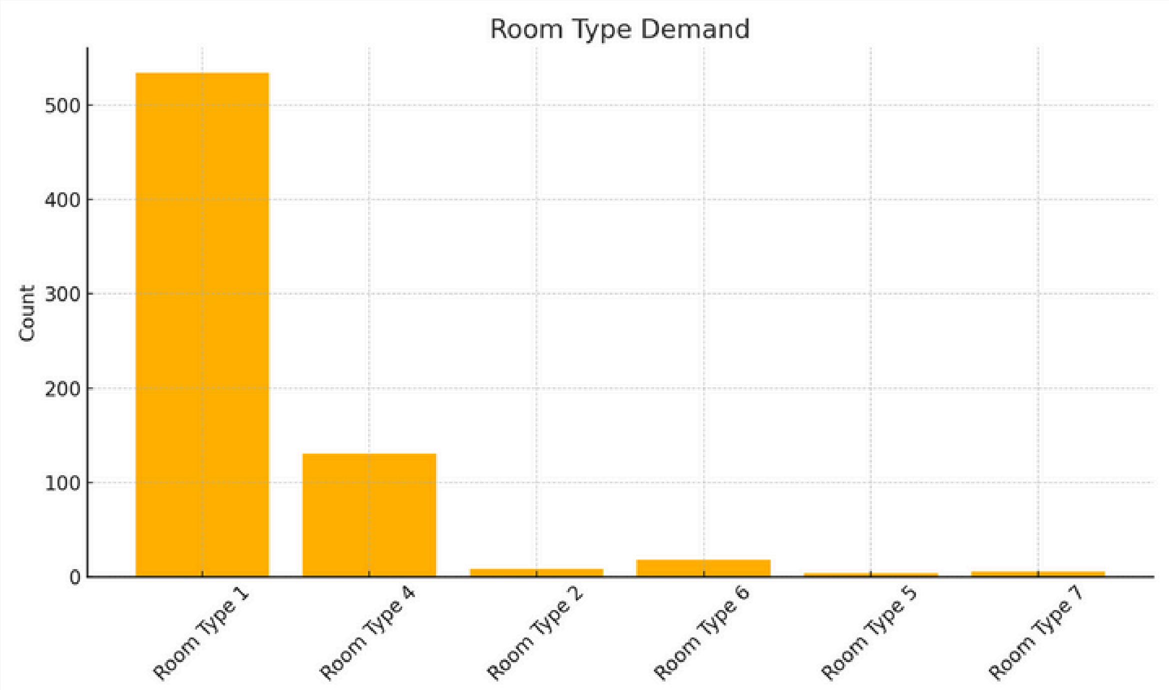
- Most Preferred:- Meal Plan 1
- Least preferred:- Meal Plan 2

2. Type of room booked:-

```sql
#Q5
Select room_type_reserved ,
count(Booking_ID)
as 'No. of Reservation '
from hotel.hotel_reservation_dataset
group by room_type_reserved;
```

Result Grid | Filter Rows: | Export:

room_type_reserved	No. of Reservation
Room_Type 1	534
Room_Type 4	130
Room_Type 2	8
Room_Type 6	18
Room_Type 5	4
Room_Type 7	6



Insights from the chart:-

- Most Booked:- Room Type 1
- Less Booked:- Rooms 2,5,6,7 are very less preferred
- While Room Type 4 is moderately preferred

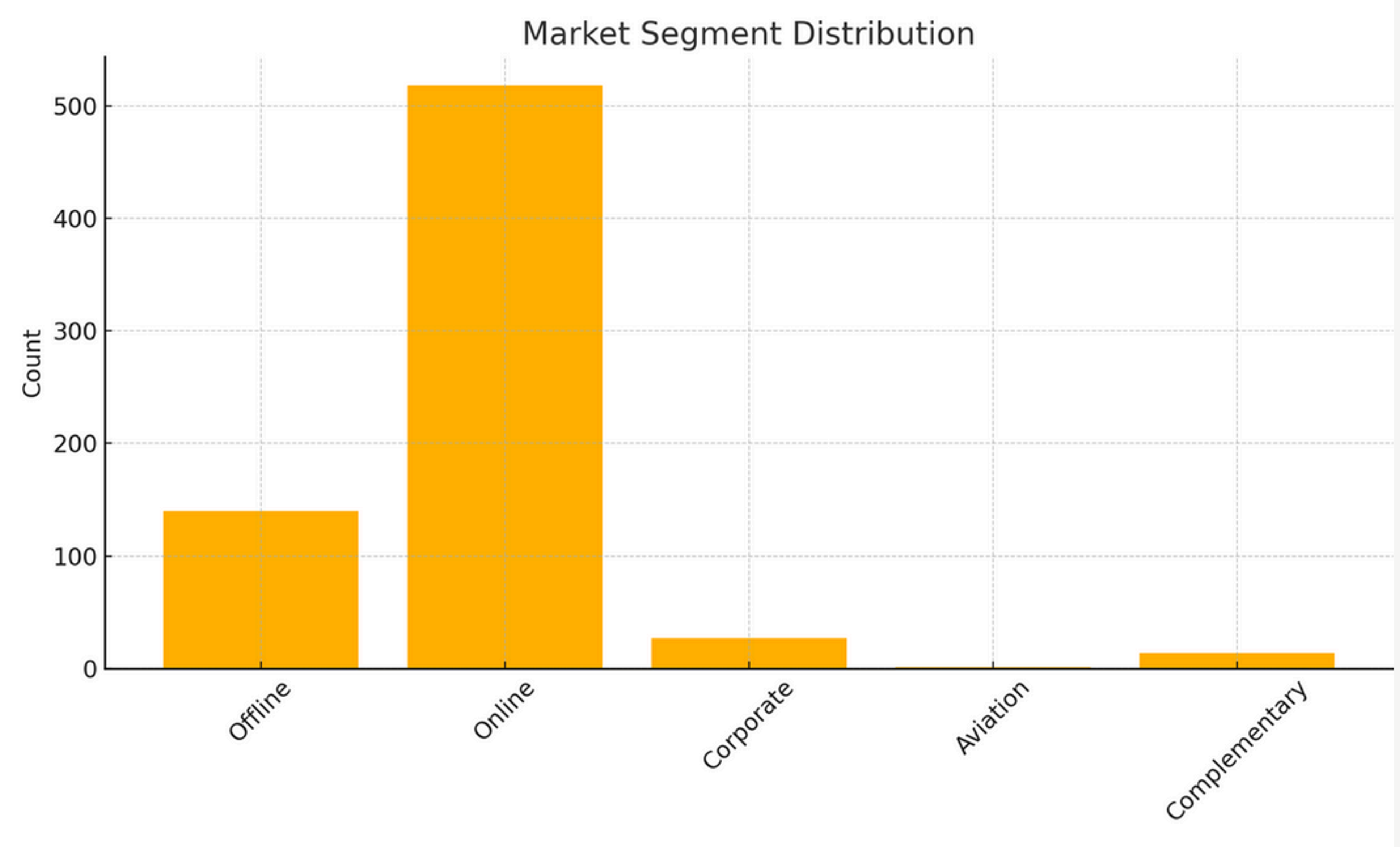
Exploratory Data Analysis

3. Market Segment Distribution:-

```sql
#Q8
Select market_segment_type ,
count(Booking_ID)
from hotel.hotel_reservation_dataset
group by market_segment_type;
```

Result Grid | Filter Rows: | Export:

market_segment_type	count(Booking_ID)
Offline	140
Online	518
Corporate	27
Aviation	1
Complementary	14



Insights from the chart:-

- Customers prefer Online Bookings
- Whereas customers show moderate interest in Offline Bookings
- Corporate, Aviation, and Complementary segments have minimal contribution

Exploratory Data Analysis

4. Month-wise Reservation Counts:-

```
#Q12
Select month(date)
as 'Month_of_Year' ,
count(Booking_ID)
as 'No. of Reservation'
from hotel.hotel_reservation_dataset
group by month(date);
```

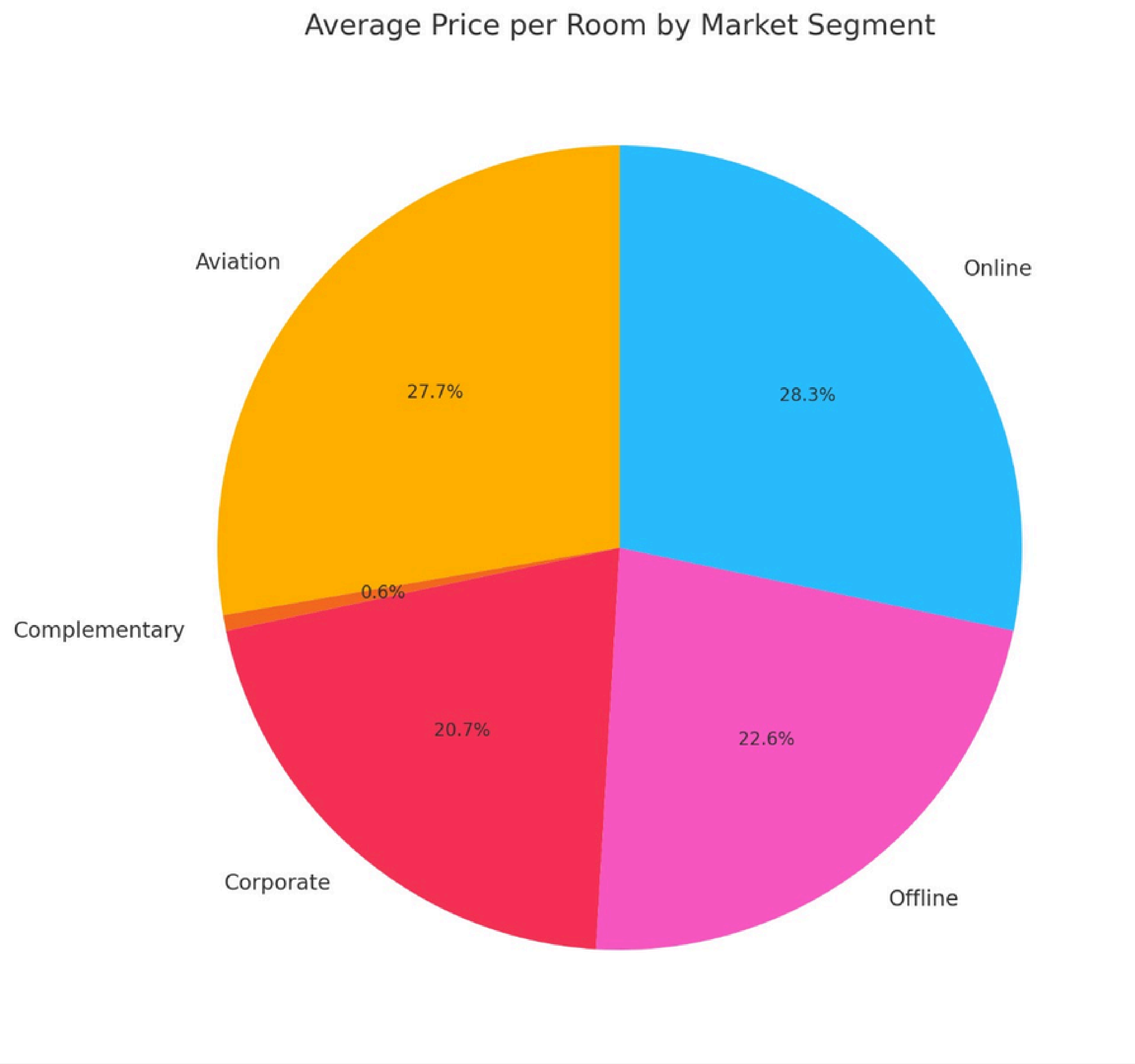
Month_of_Year	No. of Reservation
2	43
6	87
5	61
11	50
9	67
10	95
12	42
4	48
3	60
7	42
1	29
8	76

Insights from the chart:-

- Peak demands in months like June, August, and October
- January is the least preferred month by the customers
- Third Quarter (Jul-Sep) shows peak demand, maybe due to the monsoon or travel season

Exploratory Data Analysis

5. Average Pricing of each room




```sql
#Q15
Select market_segment_type,
avg(avg_price_per_room)
as 'Average Room Price'
from hotel.hotel_reservation_dataset
group by market_segment_type;
```

Result Grid | Filter Rows: | Export:

market_segment_type	avg(avg_price_per_room)
Offline	89.98171428571426
Online	112.45521235521232
Corporate	82.4011111111111
Aviation	110
Complementary	2.5357142857142856

Insights from the chart:-

- On an avg., Prices for Online Bookings are the highest per room booked
- While the Offline or Corporate segments have moderate prices per room

Business Impact

1. **Meal Plan Distribution:-** Helps in optimizing menu planning and pricing based on guest preferences. They could apply offers to meal 2, like subsidised rates upon booking a room for a certain no. of days, to increase its demand.
2. **Room Type Frequency:-** Better room inventory allocation and dynamic price adjustments. During peak seasons, rates can be increased for rooms in demand (like room 1), and offers can be applied to low-demand rooms (like rooms 2,5, and 7).
3. **Market Segment Breakdown:-** More profitable channels, like online bookings, can be enhanced further by good investments through digital ads or marketing.
4. **Month-wise Reservation Counts:-** Helps in the Optimization of staff, inventory, and pricing of rooms. The hotel can lay off or hire more staff according to the demand throughout the year. During low-demand months, they can introduce discounts, packages, and targeted campaigns.
5. **Average Pricing by Booking Method:-** Helps in identifying high-value customer segments for targeted revenue strategies.

Challenges Faced & Conclusion

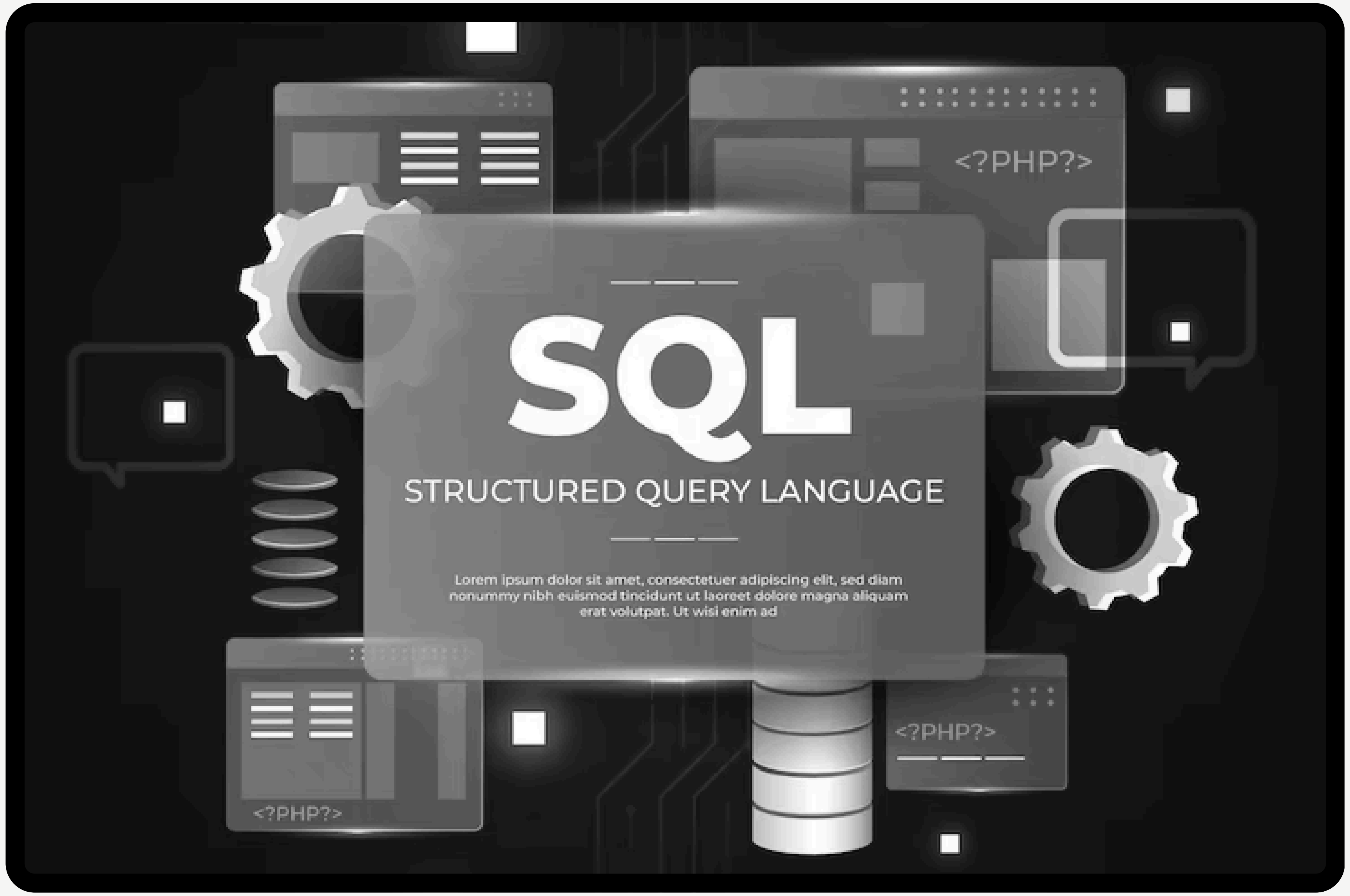
Challenges Faced

- Understanding the dataset schema and relationships.
- Data Manipulation for missing or inconsistent data for proper analysis using SQL.
- Optimizing SQL queries for grouping, filtering, and consistent EDA.
- Translating raw query outputs into business insights.
- Time management and scope expansion.

Conclusion

- Successfully performed SQL-based exploratory data analysis on 700+ hotel reservations.
- Performed EDA on various trends like Meal preference, Booking channels, etc.
- Provided actionable insights to support hotel operations, marketing, and revenue management.
- Strengthened technical skills in SQL, data manipulation and cleaning, EDA, visualization, and business interpretation

Thank you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SQL
STRUCTURED QUERY LANGUAGE
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